

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	B	D	A	A	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	B,D	B,D	A,D	A,B,C	C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	500.00	3.60	2.40	0.66 to 0.67	0.86 to 0.87	64.50 to 64.60

PART-2 : CHEMISTRY

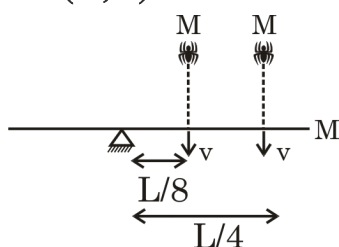
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	A	D	B	D	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,C	A,B,D	B,C,D	A,B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.50	2.16 or 2.17	2.50	5.00	2.00	13.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	A	D	D	A	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,D	A,C	A,B,C,D	A,B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.95	1.25	7.00	1.20	0.66 or 0.67	1.50

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (ii)

10. Ans (A,D)



Conserving angular momentum

$$Mv \frac{L}{4} + Mv \frac{L}{8} = \left[\frac{ML^2}{12} + M \left(\frac{L}{8} \right)^2 + M \left(\frac{L}{4} \right)^2 \right] \omega$$

$$\omega = \frac{72V}{L}$$

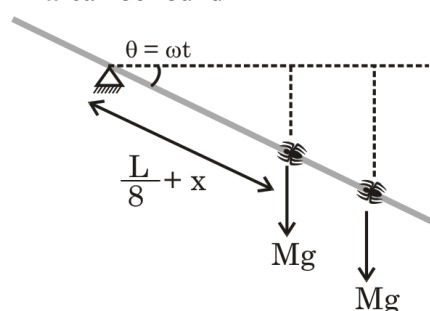

 Here ω is given constant and I varies for system of (rod + insects)

$$\tau = \omega \frac{dI}{dt}$$

$$Mg \left[\frac{L}{4} + x + \frac{L}{8} + x \right] \cos \omega t$$

$$= \omega \frac{d}{dt} \left[\frac{ML^2}{12} + M \left(\frac{L}{8} + x \right)^2 + M \left(\frac{L}{4} + x \right)^2 \right]$$

$$\Rightarrow \frac{dx}{dt} \propto \cos \omega t \Rightarrow \int dx \propto \int \cos \omega t dt$$

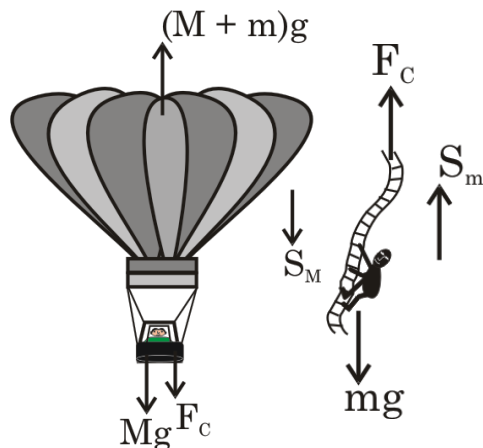
 $\Rightarrow \omega$ can be found


PART-1 : PHYSICS

SECTION-II

1. Ans (500.00)

Individual free body diagrams



$$F_{\text{net}} = 0 \text{ on } (M + m)$$

$$\Rightarrow a_{\text{cm}} = 0 \Rightarrow \Delta v_{\text{cm}} = 0 \text{ (here } v_{\text{cm}} = 0)$$

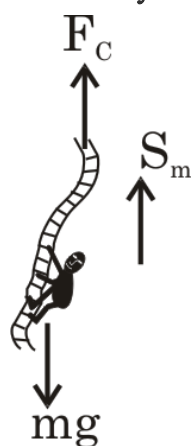
$$\Rightarrow \Delta y_{\text{cm}} = 0 \Rightarrow ms_m + M(-s_m) = 0$$

$$\Rightarrow s_m : s_M = M : m$$

$$\text{Also } s_M + s_m = h \Rightarrow s_m = \frac{Mh}{M+m} \uparrow$$

$$\Rightarrow s_M = \frac{mh}{M+m} \downarrow$$

Work-energy analysis from ground frame
on man as system w.r.t. ground



$$W_{F_c \text{ on hand}} + W_{mg} + W_{\text{int F of mass}} = 0$$

$$-mgS_M - mgS_m$$

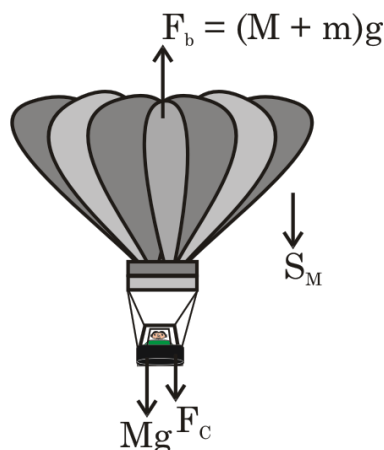
On ladder + Balloon as system w.r.t. ground

$$(i) W_{F_c \text{ on balloon}} + W_{Mg} + W_{F_b} = 0$$

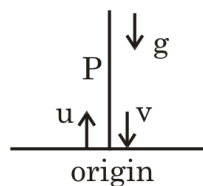
$$MgS_M - (M+m)gS_M$$

$$s_m = \frac{Mh}{M+m} \uparrow$$

$$s_M = \frac{mh}{M+m} \downarrow \Rightarrow W_{\text{int F of man}} = mgh$$



2. Ans (3.60)



$$m\bar{g} - k\bar{v} = m\bar{a}$$

$$\int_0^T m\bar{g}dt - k \int_0^T \bar{v}dt = m \int_0^T \bar{a}dt$$

$$m\bar{g}T - k\bar{s} = m(\bar{v} - \bar{u})$$

$$|\bar{g}T| = |\bar{v} - \bar{u}|$$

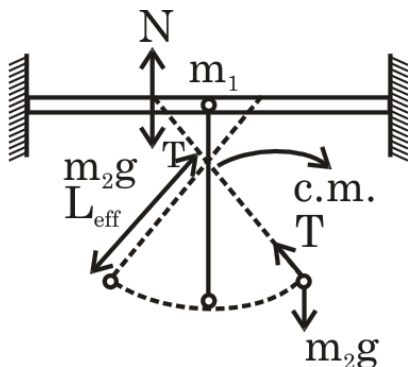
$$gT = v + u \Rightarrow T = \frac{v+u}{9}$$

$$\Rightarrow T = 3.6 \text{ sec}$$

3. Ans (2.40)

net force in horizontal direction is zero.

$$\Rightarrow a_{cm}|_x = 0$$



m_2 will oscillate like a simple pendulum

$$T = 2\pi\sqrt{\frac{L_{eff}}{g}}$$

$$\omega = \sqrt{\frac{g}{L_{eff}}} = \sqrt{\frac{g(m_1 + m_2)}{m_1 L}}$$

$$\omega = \sqrt{\frac{g}{L} \left(1 + \frac{m_1}{m_2}\right)}$$

$$\omega = 2.4 \text{ rad/s}$$

5. Ans (0.86 to 0.87)

$$2\pi\sqrt{\frac{r^3}{G\left(M + \frac{m}{\sqrt{3}}\right)}}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$(1 + 2x + 3x^2)^{15} = a_0 + a_1x + a_2x^2 + \dots + a_{30}x^{30},$$

diff. w.r.t. x

$$15(1 + 2x + 3x^2)^{14}(2 + 6x)$$

$$= a_1 + 2a_2x + \dots + 30a_{30}x^{24}$$

Put $x = 1$

$$\sum_{k=1}^{30} k \cdot a_k = 120.6^{14} = 20.6^{15}$$

2. Ans (A)

No. of 1 steps = x, No. of 2 steps = y

$$x + 2y = 7$$

$$\underbrace{11\dots1}_{7\text{times}} = \frac{7!}{7!} = 1; \underbrace{11\dots12}_{5\text{times}} = \frac{6!}{5!} = 6$$

$$11122 = \frac{5!}{3!2!} = 10$$

$$1,2,2,2 = \frac{4!}{3!} = 4$$

Total = 21 steps

Again from 9th to 16th steps

(total = 7 steps) we have

$$x + 2y = 7 \quad (\text{No. of ways} = 21)$$

$$\text{Total} = 21 \times 21 = 441$$

3. Ans (D)

$$a = 3^{r_1}7^{s_1}$$

$$b = 3^{r_2}7^{s_2}$$

$$c = 3^{r_3}7^{s_3}$$

$$d = 3^{r_4}7^{s_4}$$

To decide (r_1, r_2, r_3, r_4) .

Case-1 $\rightarrow r_1 = r_2 = r_3 = r_4 = r$

Case-2 $\rightarrow$ Exactly 3 gets r & other can be filled with 0, 1, 2, ..., r-1.

$${}^4C_{3,r}$$

Case-3 $\rightarrow$ Exactly 2 gets r, ${}^4C_{2,r^2}$

$$\Rightarrow \text{Total number of ways} = (6r^2 + 4r + 1)$$

Similarly (s_1, s_2, s_3, s_4) can be decided in $(6s^2 + 4s + 1)$ way.

$$\Rightarrow \text{Total number of quadruples} = (6r^2 + 4r + 1)(6s^2 + 4s + 1) = 67^2.$$

4. Ans (D)

Use $C_1 - 5C_3$ and $C_2 + xC_3$

and expand

$$\Delta = (1 + x^2 + y^2)^3$$

$$\Delta = \left(1 + x^2 + \frac{4}{x^2}\right)^3$$

Use $AM \geq GM$

$$\frac{x^2 + \frac{4}{x^2}}{2} \geq \left(x^2 \times \frac{4}{x^2}\right)^{\frac{1}{2}}$$

$$x^2 + \frac{4}{x^2} \geq 4$$

$$\left(1 + x^2 + \frac{4}{x^2}\right)^3 \geq 125$$

5. Ans (A)

Let

$$P = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \Rightarrow Q = \begin{bmatrix} b_1 + c_1 & a_1 + c_1 & a_1 + b_1 \\ b_2 + c_2 & a_2 + c_2 & a_2 + b_2 \\ b_3 + c_3 & a_3 + c_3 & a_3 + b_3 \end{bmatrix}$$

$$|P^{-1}Q| = \left| \frac{\text{adj}P \cdot Q}{|P|} \right| = \frac{|(\text{adj}P)Q|}{|P|^3}$$

$$(\text{adj}P)Q = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

$$\begin{bmatrix} b_1 + c_1 & a_1 + c_1 & a_1 + b_1 \\ b_2 + c_2 & a_2 + c_2 & a_2 + b_2 \\ b_3 + c_3 & a_3 + c_3 & a_3 + b_3 \end{bmatrix}$$

$$\Rightarrow |(\text{adj}P)(Q)| = \begin{vmatrix} 0 & |P| & |P| \\ |P| & 0 & |P| \\ |P| & |P| & 0 \end{vmatrix} = 2|P|^3$$

$$\Rightarrow \frac{|(\text{adj}P)(Q)|}{|P|^3} = 2$$

6. Ans (A)

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} = \begin{vmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= - \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = (1^3 + 2^3 + 3^3 - 3(1)(2)(3)) = 18$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,B,C,D)

section number of words

$$3 \text{ same} - 1 \text{ diff} \quad {}^1C_1 \times {}^5C_1 \quad {}^1C_1 \times {}^5C_1 \times \frac{4!}{3!}$$

$$3 \text{ same} - 2 \text{ same} \quad {}^2C_2 \quad \frac{4!}{2!2!}$$

$$2 \text{ same} - 1 \text{ diff} \quad {}^2C_1 \times {}^5C_2 \quad {}^2C_1 \times {}^5C_2 \times \frac{4!}{2!}$$

$$4 \text{ diff} \quad {}^6C_4 \quad {}^6C_4 \times 4!$$

$$\frac{4!}{4!} \quad \frac{4!}{626}$$

Words beginning with vowel

$$I \dots \Rightarrow \frac{8!}{3!}, \quad E \dots \Rightarrow \frac{8!}{3!2!}, \quad U \dots \Rightarrow \frac{8!}{3!2!}$$

$$\Rightarrow \frac{8!}{3!} \left(1 + \frac{1}{2} + \frac{1}{2} \right) = 2 \cdot \frac{8!}{3!}$$

8. Ans (A,D)

$$E_n = 3^n + {}^nC_2 3^{3n-2} + {}^nC_4 3^{n-4} + \dots$$

$$\theta_n = {}^nC_1 3^{n-1} + {}^nC_3 3^{n-3} + \dots$$

$$\therefore E_n + \theta_n = (3+1)^n; E_n - \theta_n = (3-1)^n$$

$$\therefore E_n = \frac{4^n + 2^n}{2}; \theta_n = \frac{4^n - 2^n}{2}$$

9. Ans (A,C)

$$\uparrow_x B \uparrow_y B \uparrow_z B \quad B \quad B \quad B \uparrow_w B \quad B \quad B$$

for 4 runs $\rightarrow$ select 3 places out of 7 between

B's and one end. Let x,y,z,w be number of A's

in this place $x + y + z + w = 10$,

$$x, y, z, w \geq 1 \Rightarrow {}^7C_3$$

$$\Rightarrow \text{Required arr.} = {}^2C_1 \cdot {}^7C_3 \cdot {}^9C_3.$$

10. Ans (A,B,C,D)

$$\begin{vmatrix} 2\alpha & 1 & 1 \\ \alpha & 1 & 2 \\ \beta^2 & 1 & -3\alpha \end{vmatrix} = 0$$

$$\Rightarrow 2\alpha(-3\alpha - 2) - \alpha(-3\alpha - 1) + \beta^2 = 0$$

$$\Rightarrow -3\alpha^2 + \beta^2 - 3\alpha = 0$$

11. Ans (A,B,C,D)

$$S(p, q, r) = (2p + q)^7 \left(r^2 - \frac{2}{r} \right)^9$$

$$\text{coefficient of } p^4 q^3 = {}^7C_3 2^4 \cdot {}^9C_6 (-2)^6$$

$$= 2^{12} 3 \cdot 5 \cdot 7^2$$

$$\text{Number of terms} = (7 + 1)(9 + 1) = 80$$

$$\text{Sum of coefficients} = 37 \cdot (-1)^9 = -2187$$

Numerically greatest coefficient

= Product of numerically greatest

coefficient in $f(p, q)$ and $h(r)$

$$= {}^7C_2 \cdot 2^5 \cdot {}^9C_6 (-2)^6 = 2^{13} 3^2 7^2$$

12. Ans (A,B,C)

$$10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$$

$$(A) \text{ Total divisors} = 9 \times 5 \times 3 \times 2 = 270$$

$$(B) 10! = 5!(2^5 3^3 5^1 7^1)$$

$$\Rightarrow \text{Ans} = (6)(4)(2)(2) = 96$$

$$(C) \text{ Sum} = (3^0 + 3^1 + 3^2 + 3^3 + 3^4)(5^0 + 5^1 + 5^2)(7^0 + 7^1)$$

$$= 30008$$

$$(D) 2^a \cdot 3^b \cdot 5^c \cdot 7^d$$

a can be 0, 3, 6

b can be 0, 3

c must be 0

d must be 0

$$\Rightarrow 3 \times 2 \times 1 \times 1 = 6$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.95)

$$\sum_{r=0}^{20} (r+1)^{20} C_r \cdot {}^{30}C_r$$

$$= \sum_{r=0}^{20} {}^{20}C_r \cdot {}^{30}C_r + \sum_{r=1}^{20} r \cdot \frac{20}{r} \cdot {}^{19}C_{r-1} \cdot {}^{30}C_r$$

$$(1+x)^{20} = {}^{20}C_0 x^0 + {}^{20}C_1 x^1 + \dots$$

$$(1+x)^{30} = {}^{30}C_0 x^{30} + {}^{30}C_1 x^{29} + \dots$$

$$\text{Coeff. of } x^{30} \text{ in } (1+x)^{50} = {}^{50}C_{30}$$

$$(1+x)^{19} = {}^{19}C_0 + {}^{19}C_1 x + {}^{19}C_2 x^2 + \dots$$

$$(1+x)^{30} = {}^{30}C_0 x^{30} + {}^{30}C_1 x^{29} + {}^{30}C_2 x^{28} + \dots$$

$$\text{Coeff. of } x^{29} \text{ in } (1+x)^{49} = {}^{49}C_{29} = {}^{49}C_{20}$$

$$\text{Required sum} = {}^{50}C_{30} + 20 \cdot {}^{49}C_{20}$$

$$\Rightarrow q = 20$$

$$\Rightarrow \frac{q}{q+1} = 0.952$$

2. Ans (1.25)

$$\text{Case -I : If } A = 0 \text{ then number of ways} = {}^{20}C_1$$

$$\text{Case -II : If } A = 2, \text{ then number of ways} = {}^{18}C_1$$

:

$$\text{Case X : If } A = 18 \text{ then number of ways} = {}^2C_1$$

$$\therefore \text{Total number of ways} = {}^{20}C_1 + {}^{18}C_1 + \dots + {}^2C_1$$

$$\text{Hence, } K = 110$$

3. Ans (7.00)

Required number of ways

= Total when all A's separated

– Total when A's separated and H's are together

$$= \frac{7!}{2!} ({}^8C_4) - 6! ({}^7C_4)$$

$$= \frac{7! \cdot 6!}{4! \cdot 3!} (6) = 4^1 \cdot 5^2 \cdot 6^3 \cdot 7^1$$

4. **Ans (1.20)**

For non-trivial solution

$$D = \begin{vmatrix} 1 & 1 & -2 \\ 2 & 1 & -c \\ -b & 2b & -c \end{vmatrix} = 0$$

$$\text{i.e. } 3bc + c - 10b = 0$$

$$c = \frac{10b}{3b+1}$$

$$\frac{10b}{3b+1} < 2 \Rightarrow \frac{10b - 6b - 2}{3b+1} < 0$$

$$\Rightarrow \frac{2(2b-1)}{3b+1} < 0, b \in \left(-\frac{1}{3}, \frac{1}{2}\right)$$

$$m = -\frac{1}{3}, n = \frac{1}{2}$$

$$\therefore n - m = \frac{5}{6}$$

5. **Ans (0.66 or 0.67)**

$$D = D_x = D_y = D_z = 0$$

$$(1-\lambda)(\lambda^2 + 3\lambda - 5) - 3(-\lambda - 15) - 4(1 + 9 + 3\lambda) = 0$$

$$-\lambda^3 - 2\lambda^2 - \lambda = 0$$

$$\lambda(\lambda^2 + 2\lambda + 1) = 0$$

$$\lambda = 0, -1$$

$$x = 2$$

6. **Ans (1.50)**

$$B^T B^{-1} = A \Rightarrow |A| = |B^T| |B^{-1}|$$

$$|A| = 1$$

take adjoint both sides

$$\text{adj}(B^{-1}) \cdot \text{adj}(B^T) = \text{adj}A$$

$$\text{adj}B)^{-1}(\text{adj}B)^T = B$$

$$(\text{adj}B)^T = (\text{adj}B)(B)$$

$$\Rightarrow (\text{adj}B)^T = |B|I$$

$$\text{adj}B = |B|I$$

$$\text{adj}(\text{adj}B) = |A|^{n-1}I$$

$$|A|^{(n-2)}A = |A|^{n-1}I \Rightarrow A = |A|I$$

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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,C,D	B,C	A,B	A,C,D	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.50	12.25	0.80	1.00	0.66 to 0.67	0.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	7	4	6	1	6

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,B,C	D	B,D	B,C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	12.16 or 12.17	1.33 or 1.34	4.50	4.85	11.33	3.60
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	6	6	2	8	4

PART-3 : MATHEMATICS

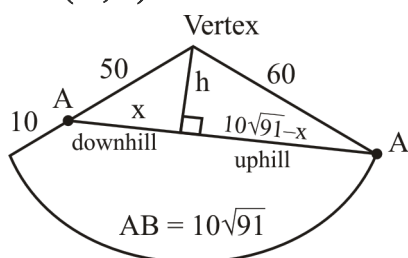
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,B,D	A,C,D	A,B,D	A,B,C	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.00	6.00	0.75	8.00	1405.00	1.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	7	1	3	6	6

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

4. Ans (A,B)



By the Pythagorean theorem

$$(10\sqrt{91} - x)^2 + h^2 = 60^2$$

$$-x^2 + h^2 = 50^2$$

$$9100 - 2(10\sqrt{91})x = 60^2 - 50^2$$

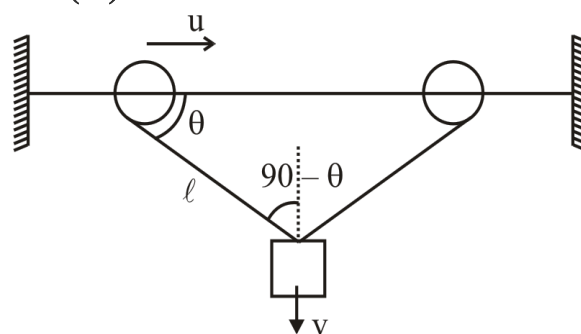
$$(20\sqrt{91})x = 8000$$

$$x = \frac{400}{\sqrt{91}}$$

PART-1 : PHYSICS

SECTION-III

6. Ans (6)



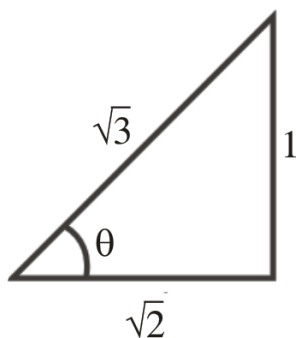
$$u \cos \theta = v \sin \theta$$

$$u = v \tan \theta$$

$$2Mgl \sin\theta = \frac{1}{2} \times 2Mv^2 + \frac{1}{2}Mu^2_x = Mv^2 \sec^2\theta$$

$$v = \sqrt{2gl \sin \theta \cos^2 \theta}$$

$$\frac{d}{d\theta}(\sin \theta \cos^2 \theta) = \cos^3 \theta + \sin \theta \times -2 \cos \theta \sin \theta = 0$$



$$\cos^2 \theta = 2 \sin^2 \theta$$

$$\tan \theta = \frac{1}{\sqrt{2}}$$

$$v = 5 \times 2^{1/4} \simeq 6$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,C)**

$$\begin{aligned} \frac{{}^nC_r}{{}^{n+4}C_4} &= \frac{n!}{r!(n-r)!} \frac{4!r!}{(r+4)!} \\ &= \frac{4!}{(n+1)(n+2)(n+3)(n+4)} ({}^{n+4}C_{r+4}) \\ &= \frac{{}^{n+4}C_{r+4}}{{}^{n+4}C_4} \end{aligned}$$

Thus,

$$\begin{aligned} S &= \frac{1}{{}^{n+4}C_4} \left[\sum_{r=0}^n 3^{r+4} ({}^{n+4}C_{r+4}) + \sum_{r=0}^3 {}^{n+4}C_r 3^r \right] \\ &= \frac{1}{{}^{n+4}C_4} \sum_{k=0}^{n+4} {}^{n+4}C_k 3^k = \frac{4^{n+4}}{{}^{n+4}C_4} \end{aligned}$$

2. **Ans (A,B,D)**

If n coplanar lines are there such that no 3 are concurrent and no two are parallel then the number of parts in which they will divide their

$$\text{plane is } 2 + 2 + 3 \dots n = \frac{n^2 + n + 2}{2}$$

$(n+1)^{\text{th}}$ plane will be intersected by previous n

planes in $\frac{n^2 + n + 2}{2}$ lines

$$\therefore f(n+1) = f(n) + \frac{n^2 + n + 2}{2}$$

$$\Rightarrow f(n) = f(n-1) + \frac{(n-1)^2 + (n-1) + 2}{2}$$

$$= 2 + \frac{1^2 + 1 + 2}{2} + \frac{2^2 + 2 + 2}{2} + \dots + \frac{(n-1)^2 + (n-1) + 2}{2}$$

$$= 2 + \frac{1}{2} \left[\frac{(n(n-1)(2n-1))}{6} + \frac{n(n-1)}{2} + 2(n-1) \right]$$

$$= \frac{n^3 + 5n + 6}{6}$$

3. **Ans (A,C,D)**

$$|A| = \begin{vmatrix} \sin \theta & \cos \theta & \tan \frac{\pi}{11} \\ \sin\left(\frac{4\pi}{3} + \theta\right) & \cos\left(\frac{4\pi}{3} + \theta\right) & \tan \frac{2\pi}{11} \\ \sin\left(\frac{8\pi}{3} + \theta\right) & \cos\left(\frac{8\pi}{3} + \theta\right) & \tan \frac{3\pi}{11} \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\Rightarrow |A| = \left(\tan \frac{\pi}{11} + \tan \frac{2\pi}{11} + \tan \frac{3\pi}{11} \right) \left(-\sin \frac{4\pi}{3} \right)$$

$$\Rightarrow |A| = \frac{\sqrt{3}}{2} \left(\tan \frac{\pi}{11} + \tan \frac{2\pi}{11} + \tan \frac{3\pi}{11} \right)$$

$$= \frac{\sqrt{3}}{2} K \text{ where } K = \tan \frac{\pi}{11} + \tan \frac{2\pi}{11} + \tan \frac{3\pi}{11}$$

$$\text{similarly } |B| = -\frac{\sqrt{3}}{2} K \text{ \& } |C| = -\frac{\sqrt{3}}{2} K$$

$$\therefore |A| + |B| = 0$$

4. **Ans (A,B,D)**

$$\text{adj}(M) = 2N, \text{adj}(N) = M$$

$$\Rightarrow |\text{adj}M| = 8|N| \text{ and } |\text{adj}N| = |M|$$

$$\Rightarrow |M|^2 = 8|N| \text{ and } |N|^2 = |M|$$

$$\Rightarrow |N| = 2$$

$$\text{Now, } MN = \text{adj}(N) \cdot M = |N|I$$

$$\Rightarrow MN = 2I \Rightarrow (B)$$

$$\text{Now } \text{adj}(M^2N) + \text{adj}(MN^2)$$

$$= \text{adj}(M \cdot 2I) + \text{adj}(2IM)$$

$$= 4(2N) + 4(M)$$

$$\text{Now, } \text{adj}(MN^{-1}) = \text{adj}(2N^{-2}) \Rightarrow (A)$$

$$= 4(\text{adj}N)^{-2} = 4M^{-2} \Rightarrow (D)$$

5. **Ans (A,B,C)**

$$(A) \frac{4!}{3!} \times \frac{4!}{2!} = 48$$

$$(B) \frac{8!}{3!2!3!} \times 2! = 1120$$

$$(C) \frac{7!}{2!2!} - \frac{6!}{2!} = 900$$

$$(D) \frac{8!}{2!2!2!} - \left[\frac{7!}{2!2!} + \frac{7!}{2!2!} - \frac{6!}{2!} \right] = 2880$$

6. **Ans (A,D)**

$$\prod_{i=1}^{99} (x+i) = A_0 + A_1x + \dots + A_{99}x^{99}$$

$$= \sum_{i=0}^{99} A_i x^i$$

Diff. w.r.t. 'x'

$$\prod_{i=1}^{99} (x+i) \cdot \sum_{i=1}^{99} \frac{1}{x+i} = \sum_{k=1}^{99} k A_k x^{k-1}$$

put $x = 1$

$$100! \sum_{i=1}^{99} \frac{1}{1+i} = \sum_{k=1}^{99} k A_k$$

$$\text{so, } r_i = 1 + i$$

$$S = \sum_{r=1}^{99} \frac{1}{\sqrt{r_k - 1}} = \sum_{i=1}^{99} \frac{1}{\sqrt{i}}$$

Use

$$2(\sqrt{n+1} - \sqrt{n}) < \frac{1}{\sqrt{n}} < 2(\sqrt{n} - \sqrt{n-1})$$

$$[S] = 18$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (5.00)**

$$\text{Let } S = 1 \cdot {}^{20}C_0^2 + \dots + 11 \cdot {}^{20}C_{20}^2$$

$$S = 11 \cdot {}^{20}C_{20}^2 + \dots + 1 \cdot {}^{20}C_0^2$$

$$2S = 12 \left[{}^{20}C_0^2 + {}^{20}C_2^2 + \dots + {}^{20}C_{20}^2 \right]$$

$$S = 6 \left[\frac{{}^{20}C_0^2 + {}^{20}C_1^2 + \dots + {}^{20}C_{20}^2}{{}^{20}C_0^2 - {}^{20}C_1^2 + \dots + {}^{20}C_{20}^2} \right]$$

$$S = 3[{}^{40}C_{20} + {}^{20}C_{10}]$$

2. **Ans (6.00)**

$$AB = BA$$

$$a = 2, b = 1$$

$$\text{Now, } AB = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} = 6I$$

$$\text{Now, } ABB^{-1} = 6B^{-1} \Rightarrow A = 6B^{-1}$$

$$\Rightarrow A^{-1}AB = 6A^{-1} \Rightarrow B = 6A^{-1}$$

$$\therefore (A+B) = 6(A^{-1} + B^{-1}) \Rightarrow \lambda = 6$$

3. **Ans (0.75)**

Number of ways = total – those in which at least three boys between the girls = 17 – a

$$\text{Now, } x G_1 y G_2 z$$

$$x + y + z = 15$$

$$\text{Min } x = 0 \text{ max } x = 12$$

$$\text{Min } y = 3 \text{ max } y = 15$$

$$\text{Min } z = 0 \text{ max } z = 12$$

Total solution = coefficient of x^{15} in

$$(x^0 + \dots + x^{12})(x^3 + x^4 + \dots + x^{15})(x^0 + \dots + x^{12})$$

$$= \dots x^{12} \text{ in } (1+x+\dots+x^{12})^3 = \dots x^{12} \text{ in}$$

$$\left(\frac{1-x^{13}}{1-x} \right)^3 = \dots x^{12} \text{ in } (1-x^{13})^3 (1-x)^{-3}$$

$$= {}^{12+3-1}C_2 = {}^{14}C_2$$

$$\therefore a = {}^{14}C_2 \times \underline{2} \times \underline{15} \text{ ways}$$

4. Ans (8.00)

$$f(n) = \sum_{r=0}^n \sum_{K=r}^n K C_r$$

$$f(n) = \sum_{r=0}^n ({}^r C_r + {}^{r+1} C_r + {}^{r+2} C_r + \dots + {}^n C_r)$$

${}^r C_r = {}^{r+1} C_{r+1}$ Put in a above equation and use formula ${}^r C_r + {}^{r+1} C_{r+1} = {}^{r+1} C_{r+1}$

$$f(n) = \sum_{r=0}^{r=n} ({}^{r+1} C_{r+1} + {}^{r+1} C_r + {}^{r+2} C_r + \dots + {}^n C_r)$$

$$= \sum_{r=0}^{r=n} {}^{n+1} C_{r+1}$$

$$= {}^{n+1} C_1 + {}^{n+1} C_2 + \dots + {}^{n+1} C_{n+1} + ({}^{n+1} C_0 - {}^{n+1} C_0)$$

$$= 2^{n+1} - {}^{n+1} C_0 = 2^{n+1} - 1$$

$$f(g) = 2^{10} - 1 = 1023 = 3 \times 11 \times 31$$

$$\text{total divisor} = (1+1)(1+1)(1+1) = 8$$

5. Ans (1405.00)

Possibilities :

$$(i) \text{ Three digit occur twice} = {}^5 C_3 \cdot \frac{6!}{2!2!2!} = 900 \text{ ways}$$

$$(ii) \text{ two digit occur three times} = 200 \text{ ways}$$

$$(iii) \text{ one digit occur four times and other twice} \\ = 300 \text{ ways}$$

$$(iv) \text{ all digits are same} = 5$$

$$\Rightarrow \text{Total possibilities} = 1405$$

6. Ans (1.00)

$$\left[\sum_{r=0}^7 \frac{(r+2)}{(r+1)} \cdot {}^7 C_r (-2)^r (3)^{7-r} \right] - 410$$

$$= \left[\sum_{r=0}^7 \left(1 + \frac{1}{r+1} \right) \cdot {}^7 C_r (-2)^r (3)^{7-r} \right] - 410$$

$$= \left[\sum_{r=0}^7 {}^7 C_r (-2)^r (3)^{7-r} + \sum_{r=0}^7 \frac{1}{r+1} \cdot {}^7 C_r (-2)^r (3)^{7-r} \right] - 410$$

$$= \left[\sum_{r=0}^7 {}^7 C_r (-2)^r (3)^{7-r} - \frac{3}{8 \times 2} \sum_{r=0}^7 {}^8 C_{r+1} (-2)^{r+1} (3)^{7-(r+1)} \right] - 410$$

$$= \left[(3-2)^7 - \frac{3}{16} \left((3-2)^8 - {}^8 C_0 (-2)^0 \cdot 3^7 \right) \right] - 410$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (3)

$$|M|^2 = k$$

$$|M|\sqrt{k} \text{ or } |M| = -\sqrt{k}$$

$$x(x-4) + x(x+6) = \sqrt{k}$$

$$2x^2 + 2x = \sqrt{k}$$

$$2x^2 + 2x - \sqrt{k} = 0$$

always two reals solutions

$$\text{or } x(x-4) + x(x+6) = -\sqrt{k}$$

$$2x^2 + 2x + \sqrt{k} = 0$$

for no solution

$$D < 0$$

$$4 - 8\sqrt{k} < 0$$

$$\frac{1}{2} < \sqrt{k}$$

$$\therefore 9k > \frac{9}{4}$$

2. Ans (7)

$$\begin{array}{ccccccc} _ & _ & _ & _ & _ & _ & _ \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \end{array}$$

$$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6$$

$$\text{where } x_2, x_3, x_4, x_5 \geq 2 \text{ and } x_1, x_6 \geq 0$$

Required number of arrangements

$$= {}^7 C_2 \times 5! = \frac{7!}{2} = \frac{k!}{2}$$

$$\text{Hence, } k = 7$$

3. Ans (1)

$$\sum_{r=1}^m \frac{(m+1)(r-1)m^{r-1}}{r^m C_r} = \sum_{r=1}^m \left(\frac{[mr - (m-r+1)]m^{r-1}}{r^m C_r} \right)$$

$$= \sum_{r=1}^m \left(\frac{m^r}{m C_r} \frac{m^{r-1} (m-r+1) r! (m-r)!}{r \cdot m!} \right)$$

$$= \sum_{r=1}^m \left(\frac{m^r}{m C_r} - \frac{m^{r-1} (m-r+1)! (r-1)!}{m!} \right)$$

$$= \sum_{r=1}^m \left(\frac{m^r}{m C_r} - \frac{m^{r-1}}{m C_{r-1}} \right) = m^m - 1$$

4. Ans (3)

$$p \neq -2 \Rightarrow y = 0 \text{ \& } x = \frac{3q-10}{q-2}, z = \frac{2}{q-2}$$

$$\therefore \alpha + \beta + \gamma = 1$$

$$\therefore \frac{3q-8}{q-2} = 1 \Rightarrow q = 3$$

5. Ans (6)

A B C No. of cases

$$\begin{array}{lcl}
 2 & 0 & 0 \rightarrow 3 \\
 2 & 1 & 1 \rightarrow 3 \\
 2 & 1 & 0 \rightarrow 6 \\
 2 & 2 & 1 \rightarrow 3 \\
 2 & 2 & 0 \rightarrow 3 \\
 2 & 2 & 2 \rightarrow 1
 \end{array} \left. \vphantom{\begin{array}{lcl} 2 & 0 & 0 \\ 2 & 1 & 1 \\ 2 & 1 & 0 \\ 2 & 2 & 1 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{array}} \right\} 19$$

Similarly for s^2 & t^2

$$\therefore \text{Ans.} = (19)^3$$

6. Ans (6)

Let $\tan A = x, \tan B = y, \tan C = z$

$$\begin{vmatrix} (y+z)^2 & x^2 & x^2 \\ y^2 & (x+z)^2 & y^2 \\ z^2 & z^2 & (x+y)^2 \end{vmatrix} = 2xyz(x+y+z)^2$$

$$= 2 \tan A \tan B \tan C (\tan A + \tan B + \tan C)^3$$

$$= 2(\tan A + \tan B + \tan C)^4 \geq 2 \times 729$$